

The Labor Market Effects of Carbon Pricing

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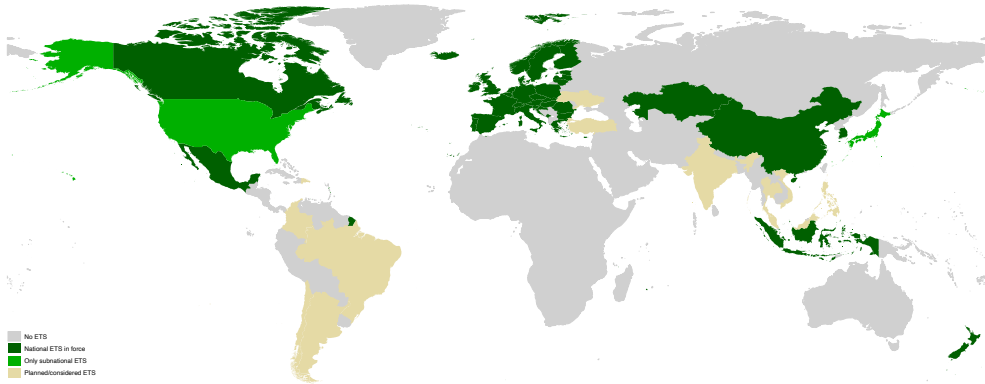
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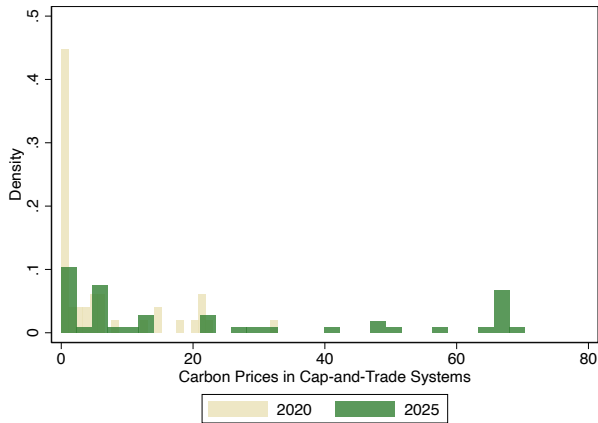
The views expressed here are those of the authors, and not necessarily those of the Norges Bank.

Cap-and-Trade Systems around the World



Cap-and-trade systems are a key carbon mitigation policy instrument

Carbon Prices in Cap-and-Trade Systems



Carbon prices are rising (though not enough to reach net-zero goals)

Carbon Pricing and the Economy

- Carbon prices are likely to further increase in the future
 - Debate about the consequences of such increases on the economy overall and on the labor market in particular (UN, ILO, EU)
- ⇒ Overall employment effects? Winners and losers?
- The design of the market
 - Worker skills

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Our paper: We study how an increase in **EU ETS carbon prices** affects **workers**

- Rich population-level data sets from the Netherlands
 - Employee-employer matched data, worker, and plant-level characteristics
- Exploit the 2017 reform in the EU ETS that reduced the supply of emission permits
- Design: Matched difference-in-differences

Overview of Findings

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Document the importance of the design of the cap-and-trade system and worker skills for labor market implications of carbon pricing

The 2017 Reform

- The carbon price until 2017 was deemed to be too low to incentivize the firms ($\approx \text{€}5$)
 - Reasons: Weak economic activity & structural oversupply (as of 2013, more than 50% of permits still freely allocated)

Phase III starts

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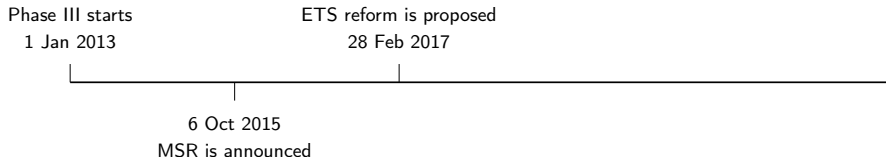
1 Jan 2013

6 Oct 2015

MSR is announced

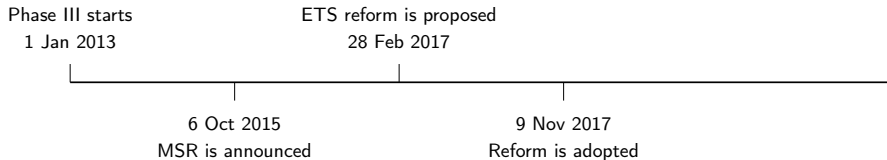
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 - Absorption of 24% of unused allowances instead of 12% if unused is above a threshold
 - Permanent cancellation of allowances



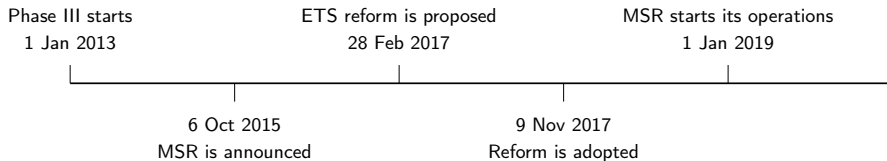
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Carbon Prices

- Stark increase in carbon prices following 2017 announcement (De Jonghe et al. 2020, Apicella & Fabiani 2023, Boeckx et al. 2025)



- In our sample, 154.6 permits per worker
- A price increase ≈ 20 euros $\rightarrow$ 3 thousand euros shock

- ETS, labor market, firm characteristics, individual characteristics
 1. ETS transactions log: Carbon emissions, free allowances (EUTL)
 2. Labor market: Wages, hours obtained from employee-employer matched data, employment status (CBS)
 3. Firm characteristics: Balance sheet, income statement, sector (CBS)
 4. Individual characteristics: Education, age, gender (CBS)
 5. Plant-level production statistics (CBS)

Empirical Strategy – Matching

- 212 ETS installations/plants belonging to 181 firms
 - ETS firms are larger and more profitable, workers are older and earn more
- ⇒ Use a matching approach to make workers in ETS and non-ETS firms more comparable

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Matching is done at the worker and firm levels. Specifically:

- Start with workers aged 25 to 60 and employed at same company in 2015 and 2016
- Match exactly on industry, gender, high-tenure dummy, all in 2016
- Within each cell, estimate propensity score of being at ETS firm, as function of:
 age_{t-1} , $\log(\text{hr. wage}_{t-1})$, $\log(\text{hr. wage}_{t-2})$, $\log(\text{firm size})_{t-1}$, and $\text{firm profit/employee}_{t-1}$
- For every treated worker, select non-ETS worker with closest propensity score, with caliper 0.025

Balance Test

- Matching removes the differences between the ETS and non-ETS groups

Variable	Full Sample			Matched Sample		
	Control	Treated	Diff	Control	Treated	Diff
Age _{t-1}	40.261 (0.105)	41.196 (0.317)	0.934 (0.333)	42.476 (0.353)	42.495 (0.268)	0.019 (0.443)
log(Hourly Wage _{t-1})	3.019 (0.008)	3.354 (0.016)	0.335 (0.018)	3.358 (0.046)	3.371 (0.030)	0.013 (0.055)
log(Hourly Wage _{t-2})	2.965 (0.009)	3.293 (0.015)	0.328 (0.018)	3.310 (0.046)	3.323 (0.029)	0.013 (0.055)
log(Size _{t-1})	5.494 (0.137)	8.375 (0.274)	2.881 (0.306)	6.426 (0.178)	6.443 (0.125)	0.017 (0.218)
Profits _{t-1} /Workers _{t-1}	20.170 (1.285)	45.403 (13.294)	25.233 (13.319)	90.652 (14.575)	86.894 (13.700)	-3.758 (19.974)
Observations	1,670,567	95,441	1,766,008	17,541	17,541	35,082

Note: all continuous variables are winsorized at 1st and 99th percentile throughout.

Exploit the increase in carbon prices in a matched difference-in-differences setting:

$$y_{it} = \beta ETS_i \times Post_t + \gamma_i + \delta_t + \epsilon_{it},$$

where:

- $ETS_i = 1$ for workers in ETS firms as of 2016, $ETS_i = 0$ for matched units
- $Post_t = 1$ if year ≥ 2018
- y_{it} : log(hourly wage), log(wages), earnings, and employment
- γ_i and δ_t are worker and year fixed effects, respectively
- Throughout, cluster standard errors at 2016-firm level

Event-study version:

$$y_{it} = \sum_{\tau=-3}^3 \beta_{\tau} ETS_i \times \mathbb{1}(t = t^* + \tau) + \gamma_i + \delta_t + \epsilon_{it},$$

where $\beta_{-1} = 0$

Baseline Results

- No adverse effects on workers – if anything, workers in ETS firms have better outcomes (in line with Dechezlepretre et al. 2023, Colmer et al. 2025, Gallé et al. 2025)
- Significant yet small positive effect on employment

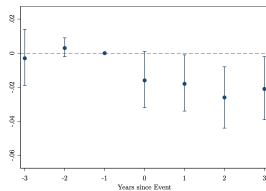
<i>Dep. Var.:</i>	Log(Hr. Wage)	Earnings	Log(Wage)	Employed
	(1)	(2)	(3)	(4)
ETS × Post	0.018 (0.016)	2,105.471 (1,482.941)	0.019 (0.017)	0.006** (0.003)
Observations	240,047	245,574	240,047	245,574
Worker FE	X	X	X	X
Year FE	X	X	X	X

Differences in Permit Shortfall/Surplus

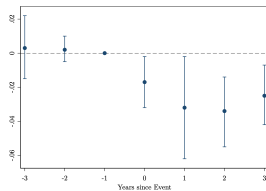
- Firms receive different amounts of free permits, depending on their industry and activity level.
 - The amount of free permits and the industries that receive more free permits are policymakers' decision
- **Permit shortfall** := emissions - free permit
- The effect of the carbon price shock is likely heterogeneous depending on the firm's permit shortfall
- Firms with larger permit shortfalls systematically uses more permits than it receives; hence, a higher price will penalize the firm more
- We normalize the difference by the number of workers in the plant

Permit Shortfall: Event-Study Evidence

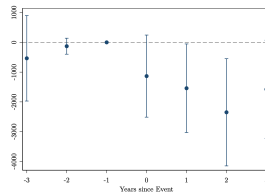
A. log(Hourly Wage)



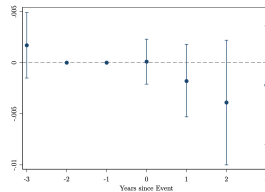
C. log(Wage)



B. Earnings



D. Employed



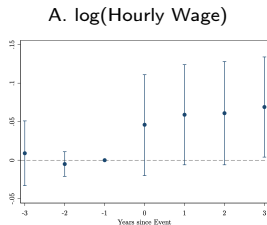
- 1 std dev higher permit shortfall $\Rightarrow$ 1.8 percent differential decline in hourly wages

Heterogeneity in Worker Skills/Education

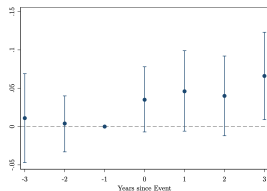
- When carbon prices rise, transition-relevant skills generate higher revenue
- Workers with such skills may capture a fraction of the higher revenue, increasing their wages
- We proxy for transition-relevant skills by STEM education
→ STEM workers are the most valuable to cut emissions (Vona et al. 2018, Saussay et al. 2022)



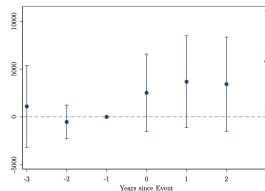
STEM Workers: Event-Study Evidence



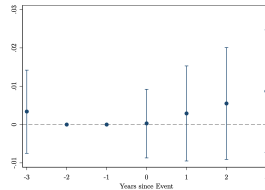
C. log(Wage)



B. Earnings



D. Employed



- **STEM workers in ETS firms has 6.9% higher hourly wages compared to STEM workers in non-ETS firms three years after the event (DDD difference = 5%)**

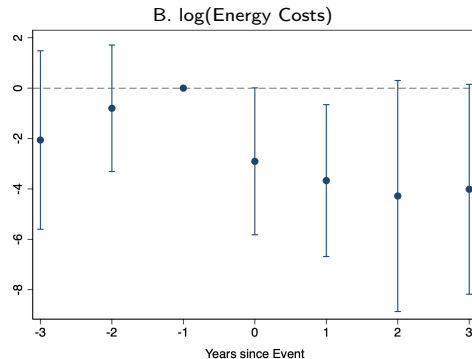
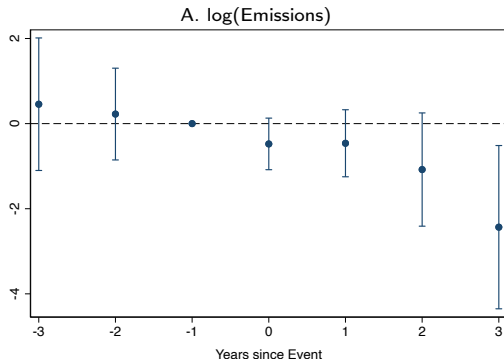
- Are the higher wage premiums for STEM workers “justified”?
- We use detailed plant-level production statistics and emission data for within-ETS and industry-year comparison:

$$\log(y_{jt}) = \beta Post_t \times \text{Frac. STEM Workers}_j + \alpha_j + \delta_{ts} + \varepsilon_{jt},$$

where y is either:

- $\log(\text{Emissions})$
- $\log(\text{Energy Costs})$
- j , t , and s index plants, years, and sectors, respectively
- We also look at the fraction of *non*-STEM college graduates as a placebo test
- Fraction of STEM workers in the data: Average=5.6%; Standard deviation=7.1%

STEM Workers, Emissions, and Energy Costs – Event-Study Evidence



- 1 st. dev. increase in Frac. STEM → 9–10% reduction in emissions, 12–18 % reduction in energy costs

- No average losses: The 2017 EU ETS reform caused no overall wage or employment declines at the worker level
- Distributional effects: Wage impacts vary—permit shortfalls hurt workers; STEM skills yield gains
- Skills matter: STEM-intensive plants reduce energy costs more after the reform
- Rising carbon prices and new ETS systems make skill development and allocation rules central for a fair transition
- Potential future work: complementarity between worker skills and firms' green investments

Thank You!

Free Allocation Formula

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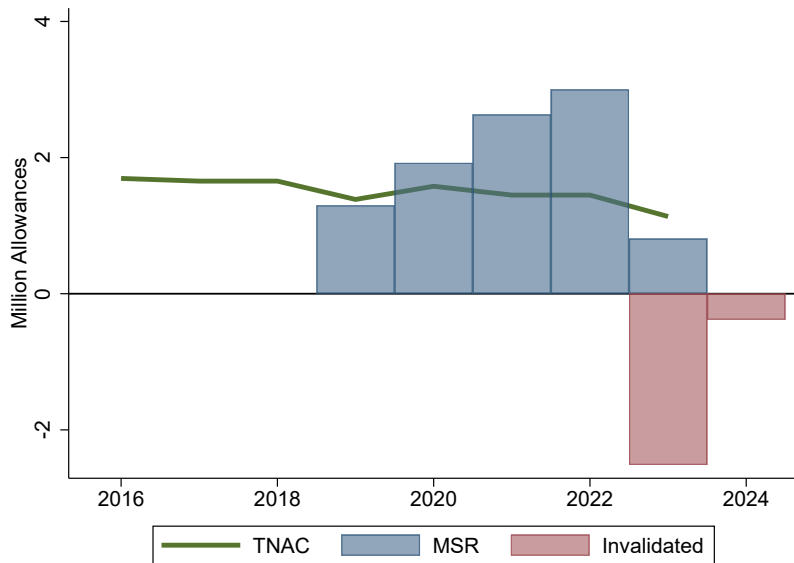
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 - Historical activity: Median activity level of the installation in earlier phases
 - Benchmark: Average emission of the best 10 percent installations in that product
 - Carbon leakage: Sectors exposed to carbon leakage receive higher free allowances
 - Linear reduction: Total allowances decrease every year

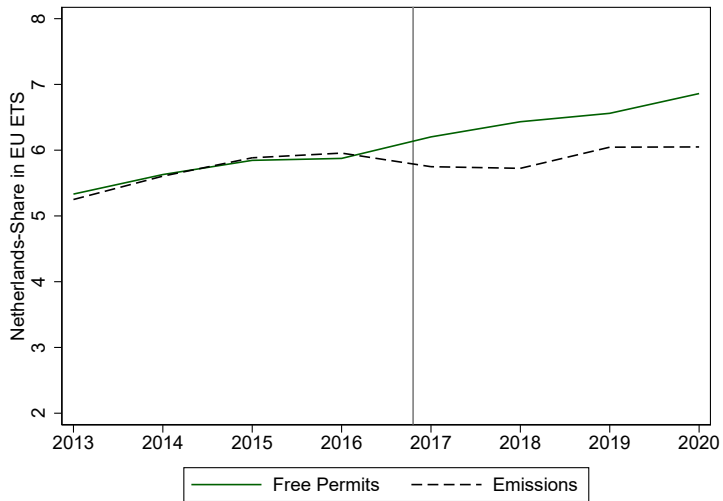


Empirical Strategy – Matching

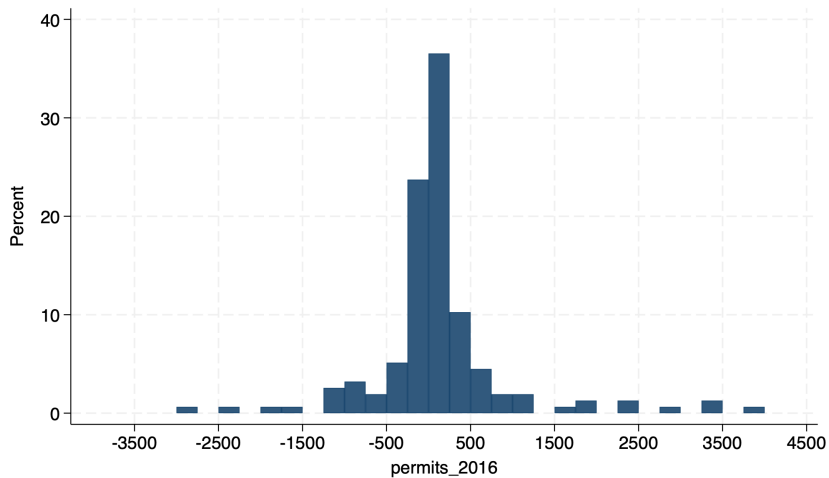
Top 10 industries with largest number of workers in ETS firms:

Number of Workers by Industry and Size Class Before Matching				
Industry	[1, 49]	[50, 249]	250+	Total
Air transport	103	133	12,414	12,650
Education	5,646	3,693	60,510	69,849
Electricity, gas, steam and air conditioning supply	488	1,229	9,947	11,664
Human health activities	16,686	6,456	109,358	132,500
Manufacture of basic metals	808	2,004	5,164	7,976
Manufacture of chemicals and chemical products	2,145	5,573	10,912	18,630
Manufacture of food products	4,934	13,365	21,367	39,666
Manufacture of motor vehicles, trailers and semi-trailers	1,107	1,945	6,032	9,084
Manufacture of other non-metallic mineral products	1,733	2,184	2,200	6,117
Manufacture of paper and paper products	804	2,829	1,875	5,508
Other	408,763	333,511	710,090	1,452,364
<u>Total</u>	443,217	372,922	949,869	1,766,008

Share of Free Permits and Emissions of Dutch Firms



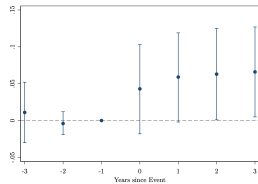
Permit Shortfall – Distribution in 2016



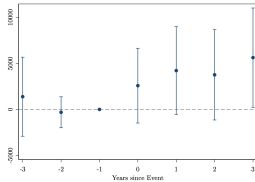
Matching with a 0.01 Caliper

Only STEM Workers

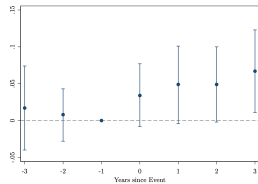
A. log(Hourly Wage)



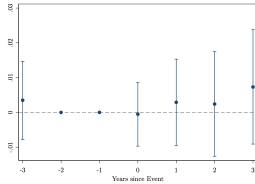
B. Earnings



C. log(Wage)

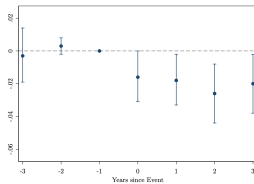


D. Employed

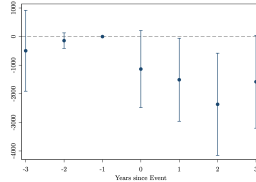


Interactions with Permit Shortfall

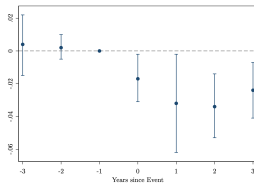
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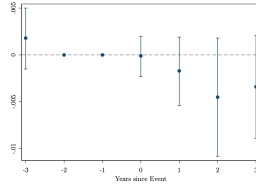
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C. log(Wage)



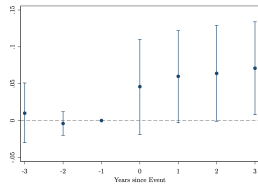
D. Employed



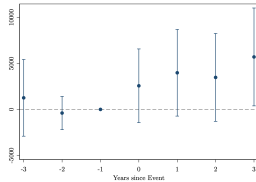
Matching with a 0.05 Caliper

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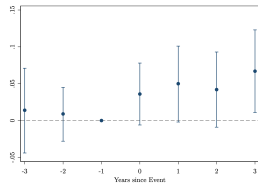
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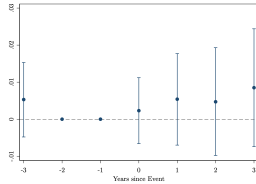
B. Earnings



C. log(Wage)

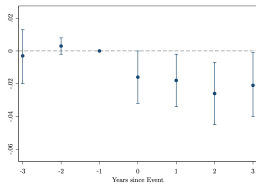


D. Employed

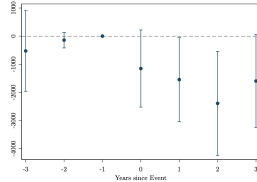


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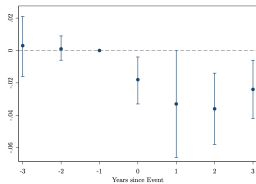
A. log(Hourly Wage)



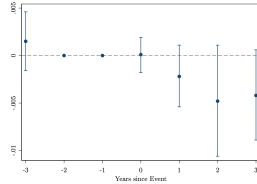
B. Earnings



C. log(Wage)



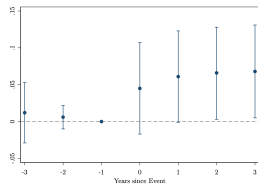
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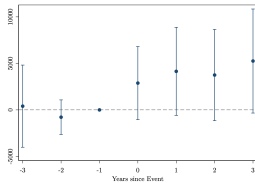
Matching on one Lag of log(Hourly Wage)

Only STEM Workers

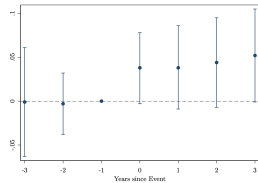
A. log(Hourly Wage)



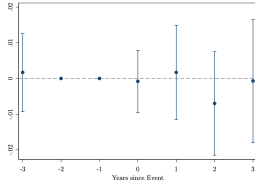
B. Earnings



C. log(Wage)

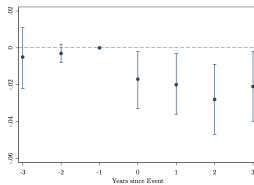


D. Employed

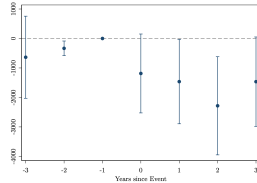


Interactions with Permit Shortfall

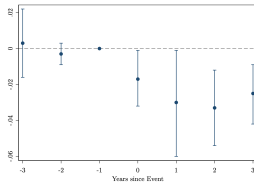
A. log(Hourly Wage)



B. Earnings



C. log(Wage)



D. Employed

